

Stat 390 Presentation 2

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Code - Tasks 1-4 & 8

- Task 1 - KimiaNet instead of DenseNet
 - DenseNet121 backbone that loads KimiaNet .pth weights
- Task 2 - Don't use RandomResizedCrop()
 - Remove patches smaller than 32x32
- Task 3 - Remove saturation jitter from dataset.py
- Task 4 - Use AdamW instead of Adam
 - Increases likelihood of finding a global minimum instead of a local minimum
- Task 8 - Set persistent workers to true
 - Added parameter and set to true in create_data_loaders() function in Main.py

Code - Task 5 (Entropy Regularization)

$$H(w) = - \sum_i w_i \log w_i$$

- High entropy -> Attention spread across many patches
- Low entropy -> Attention concentrated on a few patches
- We penalize overly sharp attention distributions
- This encourages more stable and distributed evidence aggregation
- Improvements
 - Prevents attention collapse
 - Improves gradient flow to more patches
 - Reduces reliance on single slice artifacts
 - Produces more robust slide level representations
- A larger lambda encourages smoother attention while a smaller lambda encourages more selective attention

Code - Task 5 (Entropy Regularization) cont.

- Main_task5.py
 - Imports switched from models.py & trainer.py to models_task5.py & trainer_task5.py
- Models_task5.py
 - Added detach_attn_weights flag and set equal to true, allowing the caller to choose whether returned attention tensors keep gradient links
 - Replaced unconditional .detach() logic with conditional logic across attention hierarchy levels
 - Case weights pass through conditional detach to keep top level attention consistent with patch and stain levels
 - Benefits include enabling direct optimization of attention distribution shape
 - Risks include higher memory usage and potential training instability if lambda is too strong

Code - Task 5 (trainer_task5.py)

- New configs in `__init__`
 - `attn_entropy_enable` - feature flag for safe ablation
 - `attn_entropy_lambda` - strength of regularizer vs classification loss
 - `attn_entropy_eps` - prevents divide by zero instability
- New helper functions
 - `_collect_attention_tensors` - recursive extraction over tensor/dict/list/tuple
 - `_entropy_from_weights` - normalize over last dim, clamp with epsilon, compute mean entropy
 - `_attention_entropy` - gather candidate tensors, filter empties, compute mean entropy across tensors
- `train_epoch` logic change
 - Conditional forward with attention return - needs attention tensors and gradient flow
- Regularization applied to all attention levels

Code - Task 5 Results

- Test accuracy drop from 0.6667 to 0.5000
- Test loss worsened from 1.0776 to 1.5478
- 5 of 18 cases flipped prediction, all from class 1 to class 0
- Training loss curves are almost identical in scale, so optimization capacity was not reduced
- Validation got more unstable in entropy run after epoch 3
- Attention-entropy regularization changed decision threshold behavior more than ranking behavior
- Both runs are evaluated on the final trained state, not the best validation epoch

Code - Task 6

- `Models_task6_dustin.py`
 - `__init__()`:
 - Created two sequential nets
 - V: Linear $\rightarrow$ Tanh $\rightarrow$ Dropout
 - U (gating branch): Linear $\rightarrow$ Sigmoid $\rightarrow$ Dropout
 - `forward()`:
 - Calculated gated weights
 - Used $V \cdot (0.1 + U)$
 - Avoids gradient collapse

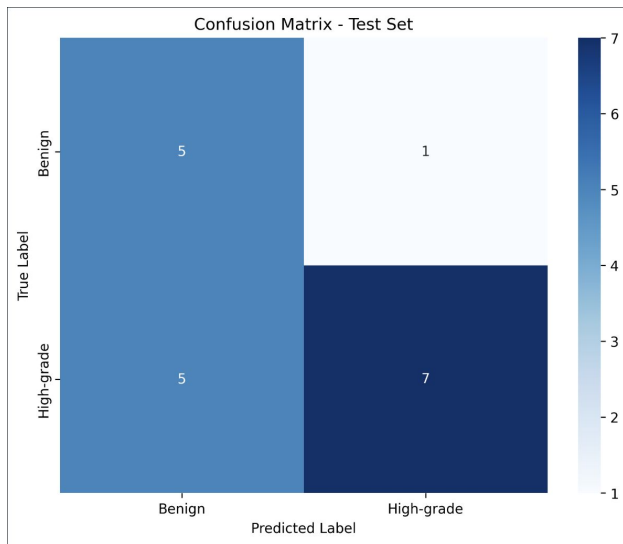
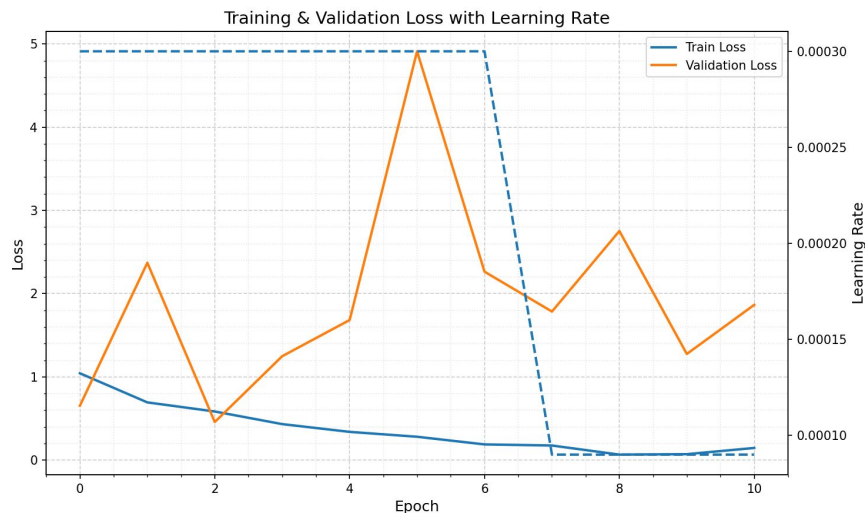
Code - Task 6 Results

- Test accuracy drop from 0.6667 to 0.3889
- Test loss worsened from 1.0776 to 6.8339
- Issues
 - Strongly over classified Benign
 - 11 high-grades misclassified as benign, no benigns misclassified as high-grades
 - Validation loss exploded after epoch 5
 - Model did not seem to learn past the first epoch (attention collapse)
- Potential solutions:
 - Change weights to $V \cdot U$
 - Consolidate entropy regularization/temperature with gating

Analysis on Tasks 1-4 & 8

Due to training issues, we were unable to perform the analysis on
Tasks 5&6 in time for this presentation.

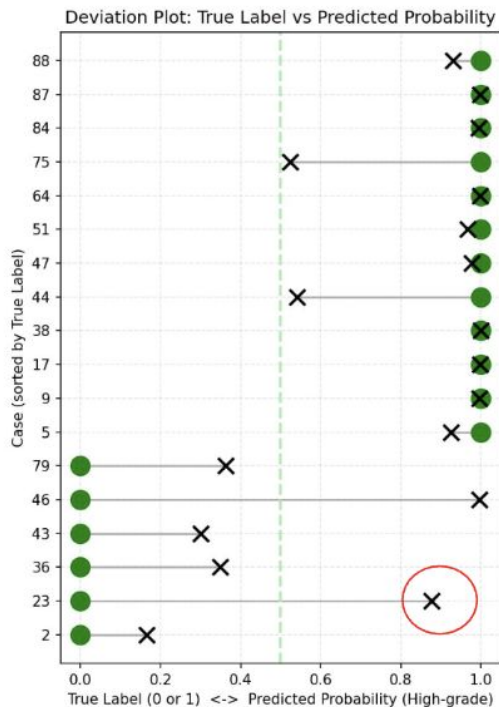
Analysis - Loss Curves & Confusion Matrix



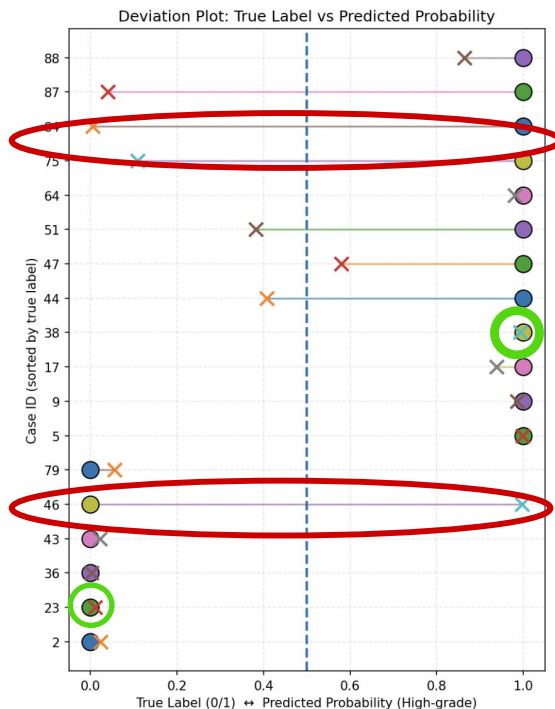
- Train loss decreases smoothly, but validation loss fluctuates and spikes → model generalization is unstable and suggests overfitting. Likely due to small # cases + high variance.
- Decent benign recall → the model has more consistently better predictions for the benign class.
- Substantial high-grade ←→ benign mistakes (false negatives) → sensitivity to high-grade class is limited.

Deviation Plot

Previously:



Currently:

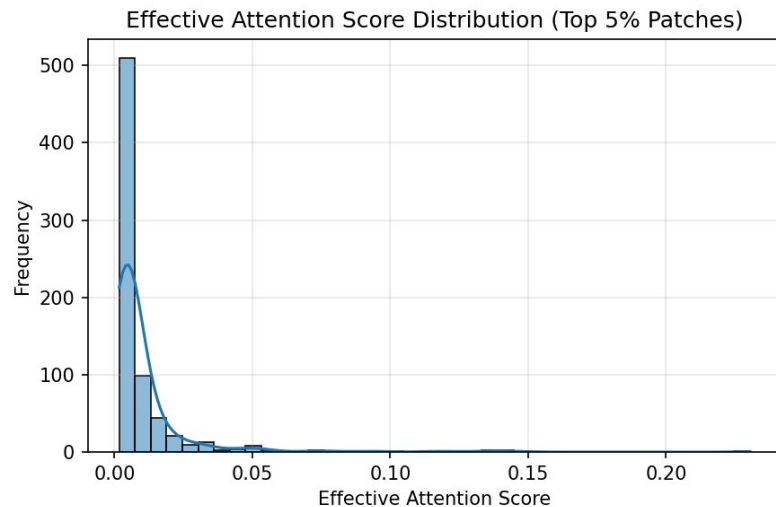
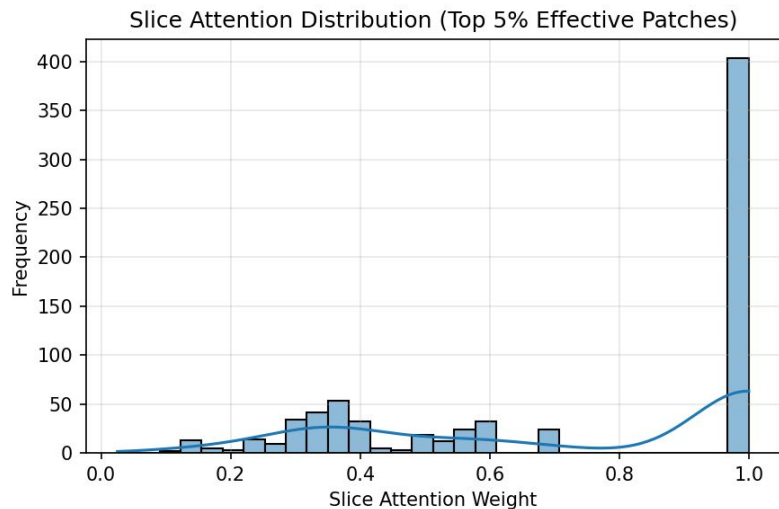


- Several cases show large gaps between true label and predicted probability, indicating miscalibration issues.
- Red-circled examples highlight cases where the model is highly confident about the wrong class.
- The current model is making errors with high certainty, which is a concern.
- For difficult cases, prediction confidence is not well aligned with correctness.

Slice & Effective Attention

Slice attention shows a few dominant regions drive predictions.

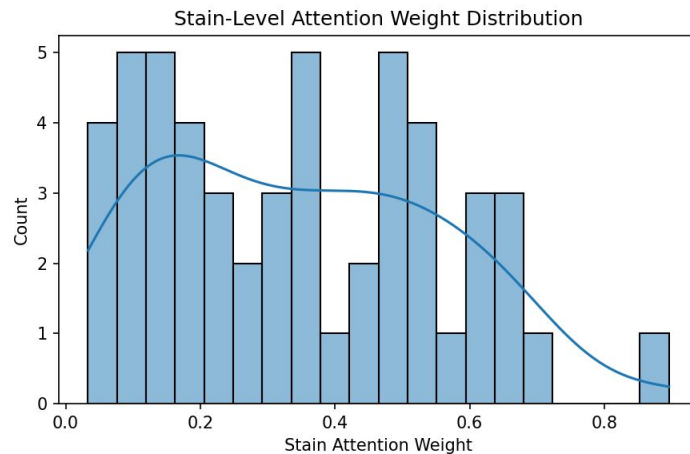
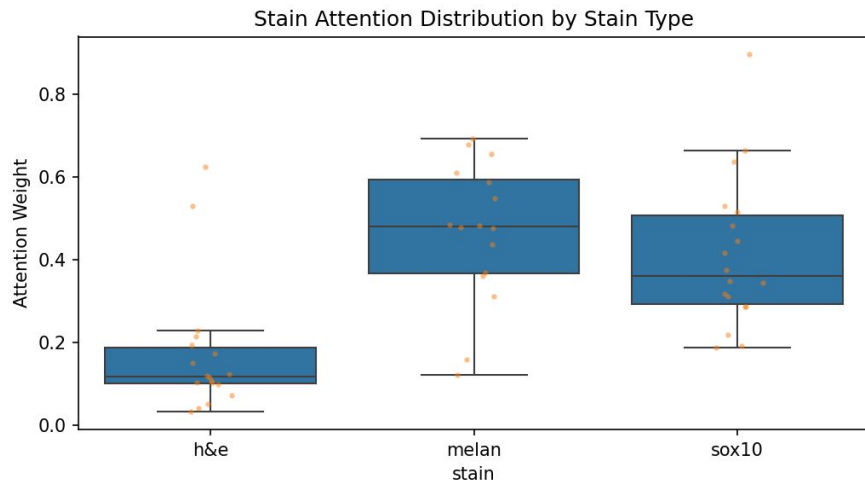
Effective attention is highly sparse, meaning only a small subset of patches influences outcomes.



Stain-Level Attention

H&E receives significantly lower attention than Melan and sox10 → stronger reliance on specific stains.

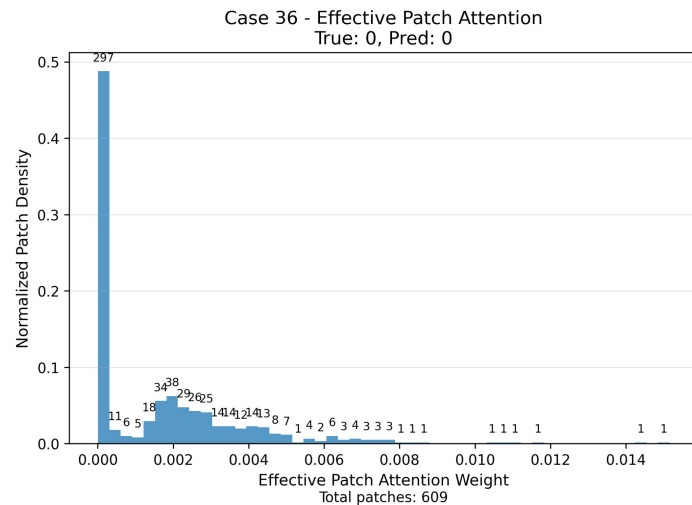
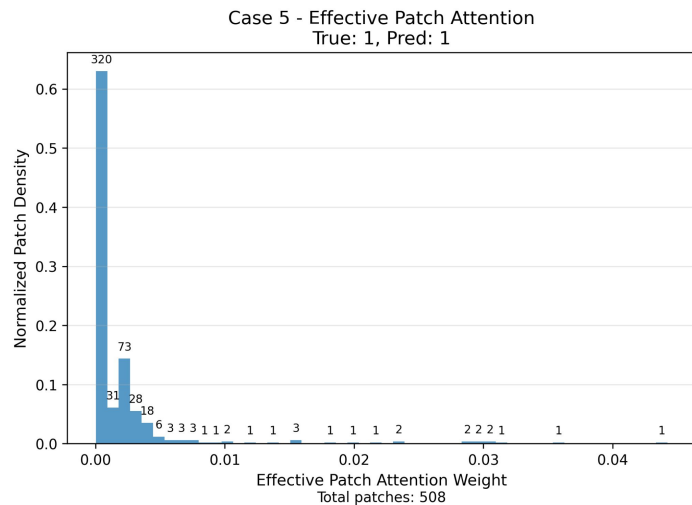
Consistent distributions indicate stable modality selection during prediction.



Correctly Classified Patches

Effective attention is sparse, with a small subset of patches driving correct predictions.

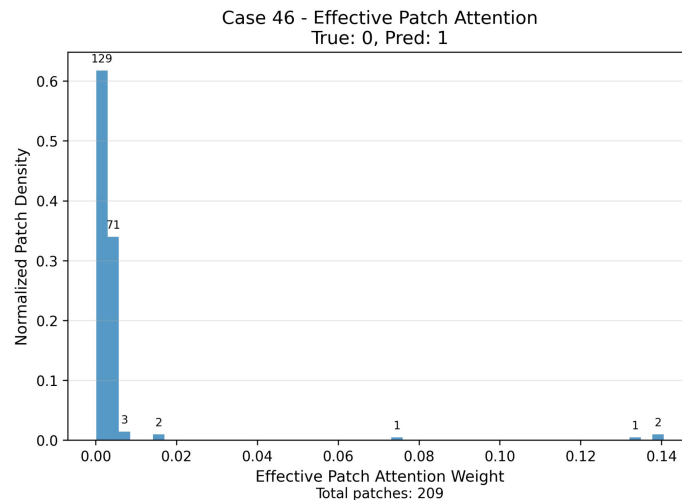
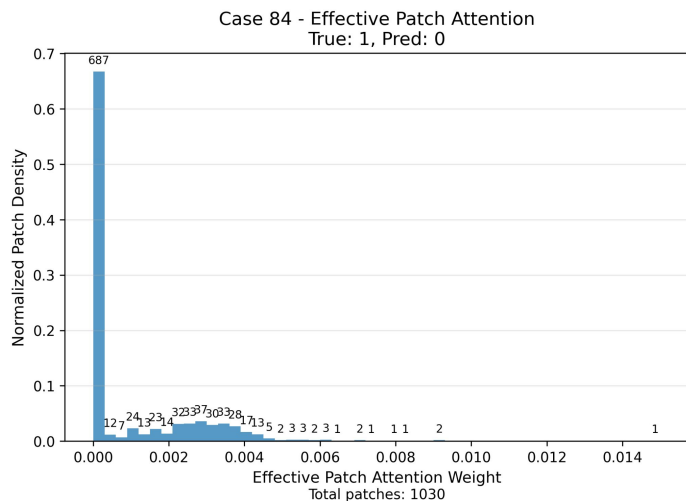
Consistent attention patterns suggest the model focuses on diagnostically meaningful regions.



Incorrectly Classified Patches

Misclassified cases depict similar sparse attention, indicating confident but misplaced focus.

Errors likely arise from ambiguous features rather than failure of the attention mechanism.



Top high attention patches for correctly classified class

Case 36 - Top Effective Patches

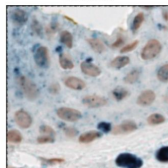
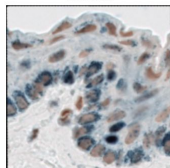
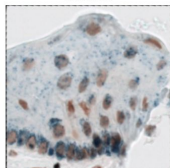
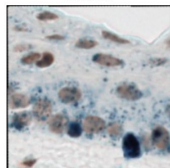
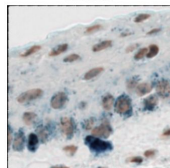
sox10 s0 p30
Eff: 0.0152

sox10 s0 p55
Eff: 0.0145

sox10 s0 p15
Eff: 0.0117

sox10 s0 p4
Eff: 0.0112

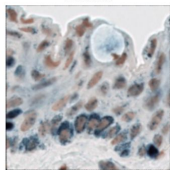
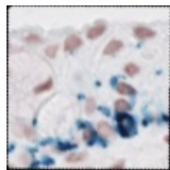
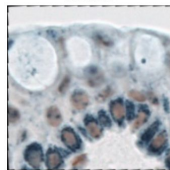
sox10 s1 p10
Eff: 0.0109



sox10 s0 p62
Eff: 0.0103

melan s1 p15
Eff: 0.0087

sox10 s0 p3
Eff: 0.0082



Benign

Case 5 - Top Effective Patches

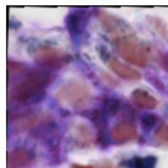
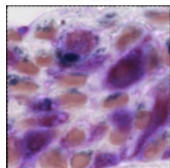
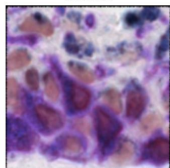
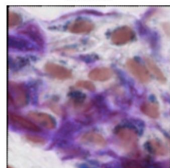
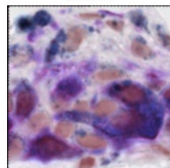
melan s0 p56
Eff: 0.0442

melan s0 p92
Eff: 0.0359

melan s0 p48
Eff: 0.0314

melan s0 p20
Eff: 0.0305

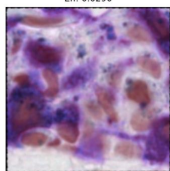
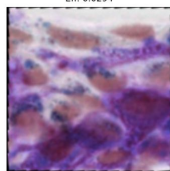
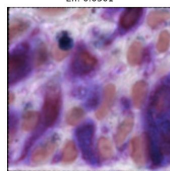
melan s0 p44
Eff: 0.0305



melan s0 p75
Eff: 0.0301

melan s0 p30
Eff: 0.0294

melan s0 p51
Eff: 0.0290

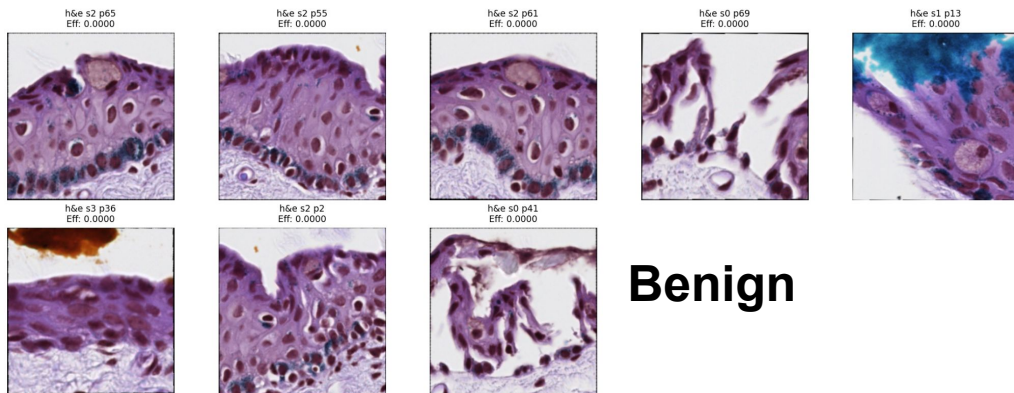


High-Grade

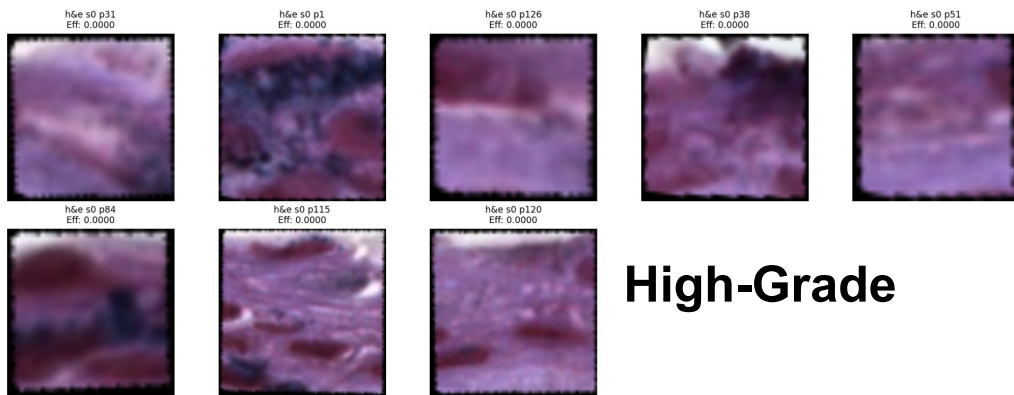
- Case 36: many patches show consistent tissue texture, some have clearer structure.
 - Sox10 → strong nuclear staining, the model may be using stain intensity or patterns.
- Case 5: patches are more cellular and dense.
 - High-attention patches appear morphologically plausible.

Top low attention patches for correctly classified class

Case 36 - Bottom Effective Patches



Case 5 - Bottom Effective Patches



- Many low-attention patches are out-of-focus, low-content- or have homogenous texture.
- Good that the model is down-weighting these non-informative patches.
- However, we must ensure that high-attention patches in misclassified cases are not also meaningless.

Top high attention patches for incorrectly classified class

Case 46 - Top Effective Patches

h&e s0 p3
Eff: 0.1405

h&e s0 p1
Eff: 0.1399

h&e s0 p22
Eff: 0.1334

h&e s0 p44
Eff: 0.0756

h&e s0 p2
Eff: 0.0162

h&e s0 p39
Eff: 0.0146

sox10 s0 p46
Eff: 0.0072

sox10 s0 p9
Eff: 0.0060

Benign

Case 84 - Top Effective Patches

melan s0 p165
Eff: 0.0150

melan s0 p30
Eff: 0.0093

melan s0 p23
Eff: 0.0092

sox10 s0 p102
Eff: 0.0082

melan s0 p98
Eff: 0.0080

sox10 s0 p45
Eff: 0.0073

melan s0 p112
Eff: 0.0071

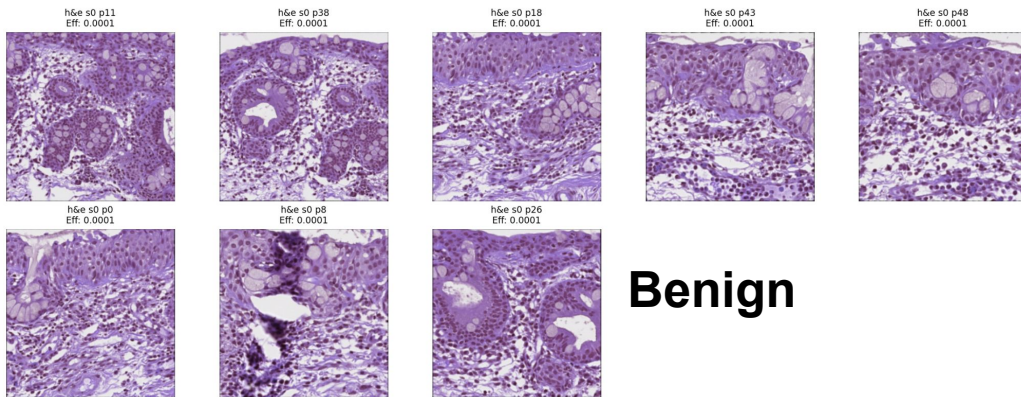
sox10 s0 p130
Eff: 0.0071

High-Grade

- Model may focus on staining artifacts or misleading textures.
- High-attention patches do not appear to be clearly diagnostic → attention being captured by ambiguous regions.
- Indicates potential domain bias or insufficient diversity in training data.
- Potentially unintended drivers of attention.

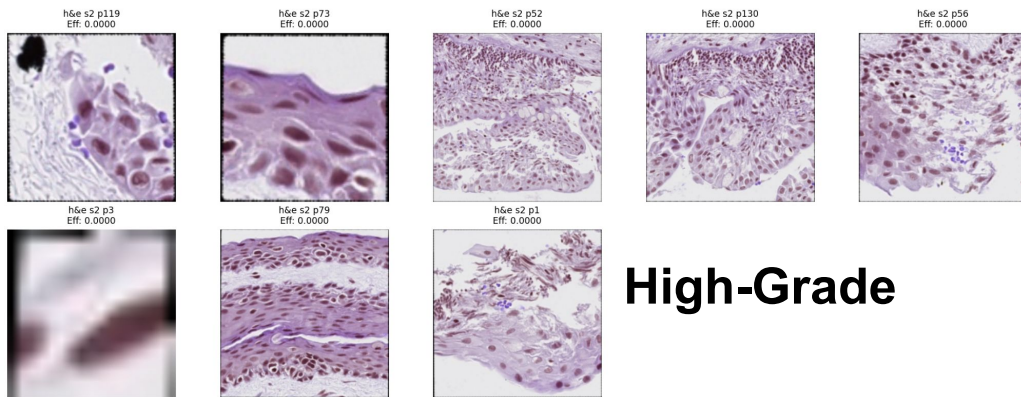
Top low attention patches for incorrectly classified class

Case 46 - Bottom Effective Patches



Benign

Case 84 - Bottom Effective Patches



High-Grade

- Low-attention patches remain mostly uninformative, even in error cases.
- Suggests the issue may be upstream, the model is assigning high attention to patches that don't reliably encode the true label → leading to confident misclassification which we saw in the deviation plot.

General Conclusions

- Generalization unstable → spiky val loss = high variance or overfitting.
- High-grade sensitivity is weak → confusion matrix shows cases that are high-grade but predicted benign (false negatives).
- Deviation plot shows confident errors → calibration problem or spurious features.
- Patch attention is skewed; slice attention piles up near 1.0 → reliance on a few instances.
- Stain-level attention distributions differ → the model may prefer certain stains in making predictions.

Next Steps

- Tune decision threshold to prioritize high-grade recall.
 - Lowering the classification threshold for high-grade will reduce false negatives (high-grade predicted benign), improving sensitivity.
- Stronger regularization, more epochs to allow for true convergence.
 - Regularization (e.g., weight decay, dropout) can reduce overfitting seen in spiky validation loss, while controlled additional epochs (with early stopping) may allow better feature stabilization instead of high-variance learning.
- Analyze the results with the implemented temperature scaling to improve calibration.
 - Temperature scaling rescales logits without changing accuracy, aligning predicted probabilities with true likelihoods — addressing the confident misclassification observed in the deviation plot.

Next Steps pt. 2

- Analyze results with entropy regularization implemented to address the attention collapse.
 - Entropy regularization encourages more distributed attention across patches, preventing extreme reliance on a few regions and improving robustness to noisy or misleading patches.
- Handle stain bias by normalizing or adding stain augmentation.
 - Stain normalization reduces domain-specific color variation, and augmentation forces the model to learn morphology instead of overfitting to stain intensity patterns.
- Patch sampling improvements to ensure each case includes enough tissue and representative regions → increase patches per bag, and tissue filtering to avoid blurs.
 - Increasing patches per bag improves coverage of diagnostically relevant regions, while filtering blurry/low-content patches reduces noise — improving both stability and generalization.

Thank you!